Practical no. 13

Aim - Implement basic Publisher – Subscriber using MQTT. Publish messages using paho mqtt. Subscribe and control LED.

Hardware Requirements

1. Raspberry Pi Model A/B/B+
2. Jumper wires (Female to Male)

- **Software Requirements**
  1. Raspbian Stretch OS
  2. MQTT Dashboard (From. Google Playstore)

1. **Connect your 4 digit 7 segment display with Raspberry Pi's GPIO Pins.**

LED pin	Function	RPI Physical Pin	Raspberry Function
VCC	POWER	12	GPIO 18
GND	GND	6	GND

- **Code**

```
import paho.mqtt.client as mqtt
import RPi.GPIO as GPIO

# ---------------- GPIO SETUP ----------------
GPIO.setmode(GPIO.BOARD)
GPIO.setwarnings(False)

# Physical pin 12
LED = 12

GPIO.setup(LED, GPIO.OUT, initial=GPIO.LOW)


# ---------------- MQTT CONNECT ----------------
def on_connect(client, userdata, flags, reason_code, properties):
    print("Connected with result code:", reason_code)

    # IMPORTANT: Same topic as your mobile app
    client.subscribe("led")

    print("Subscribed to topic: led")


# ---------------- MQTT MESSAGE ----------------
def on_message(client, userdata, msg):

    # Convert received message to lowercase
    message = msg.payload.decode("utf-8").strip().lower()

    print("Received:", message)

    # LED ON
    if message == "on":
        GPIO.output(LED, GPIO.HIGH)
        print("LED ON")

    # LED OFF
    elif message == "off":
        GPIO.output(LED, GPIO.LOW)
        print("LED OFF")

    # Unknown message
```

```
    # Unknown message
    else:
        print("Unknown message:", message)


# ---------------- MQTT CLIENT ----------------
client = mqtt.Client(
    callback_api_version=mqtt.CallbackAPIVersion.VERSION2
)

client.on_connect = on_connect
client.on_message = on_message

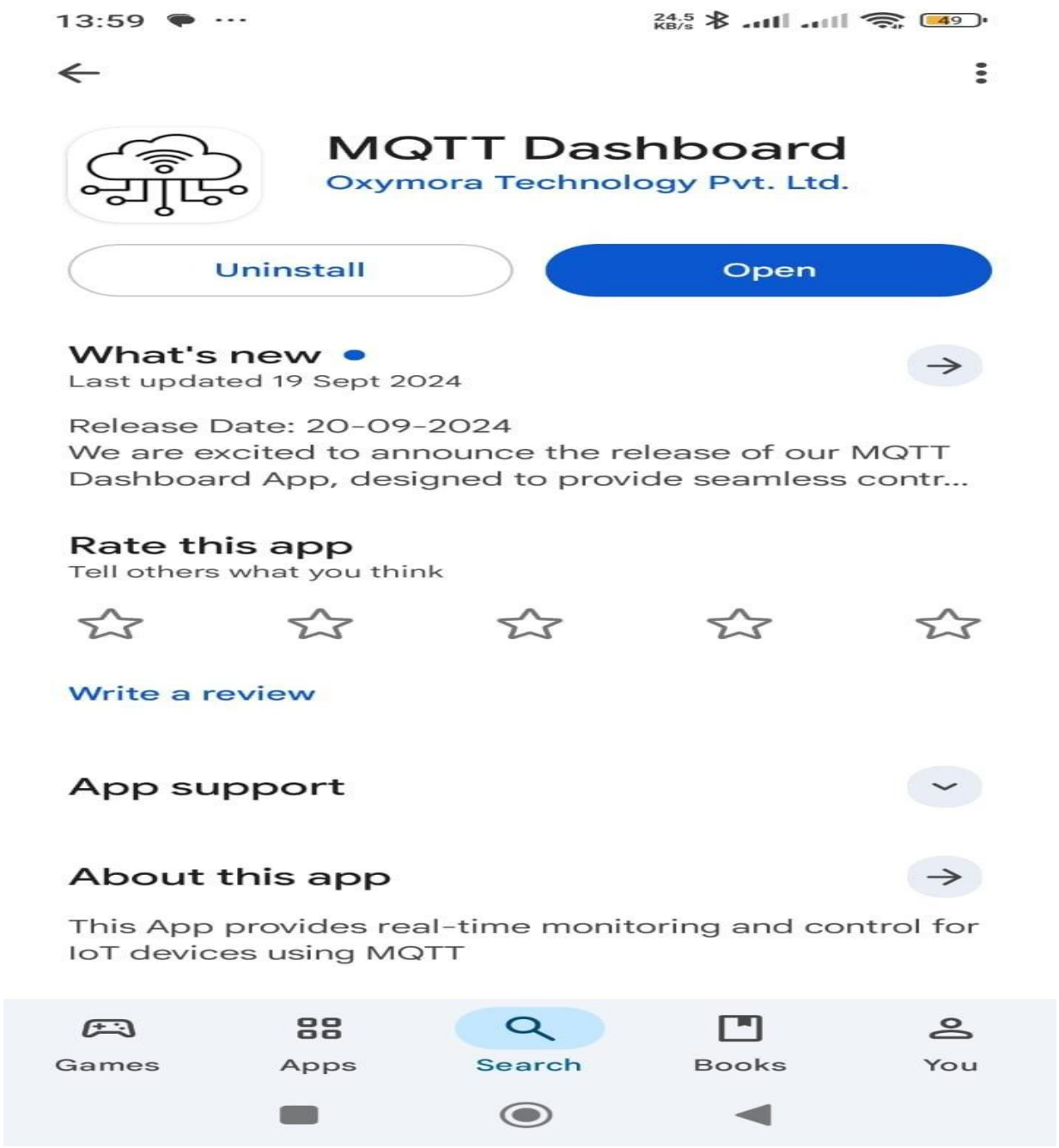

# Connect to MQTT broker
client.connect("broker.emqx.io", 1883, 60)

print("Waiting for MQTT messages...")

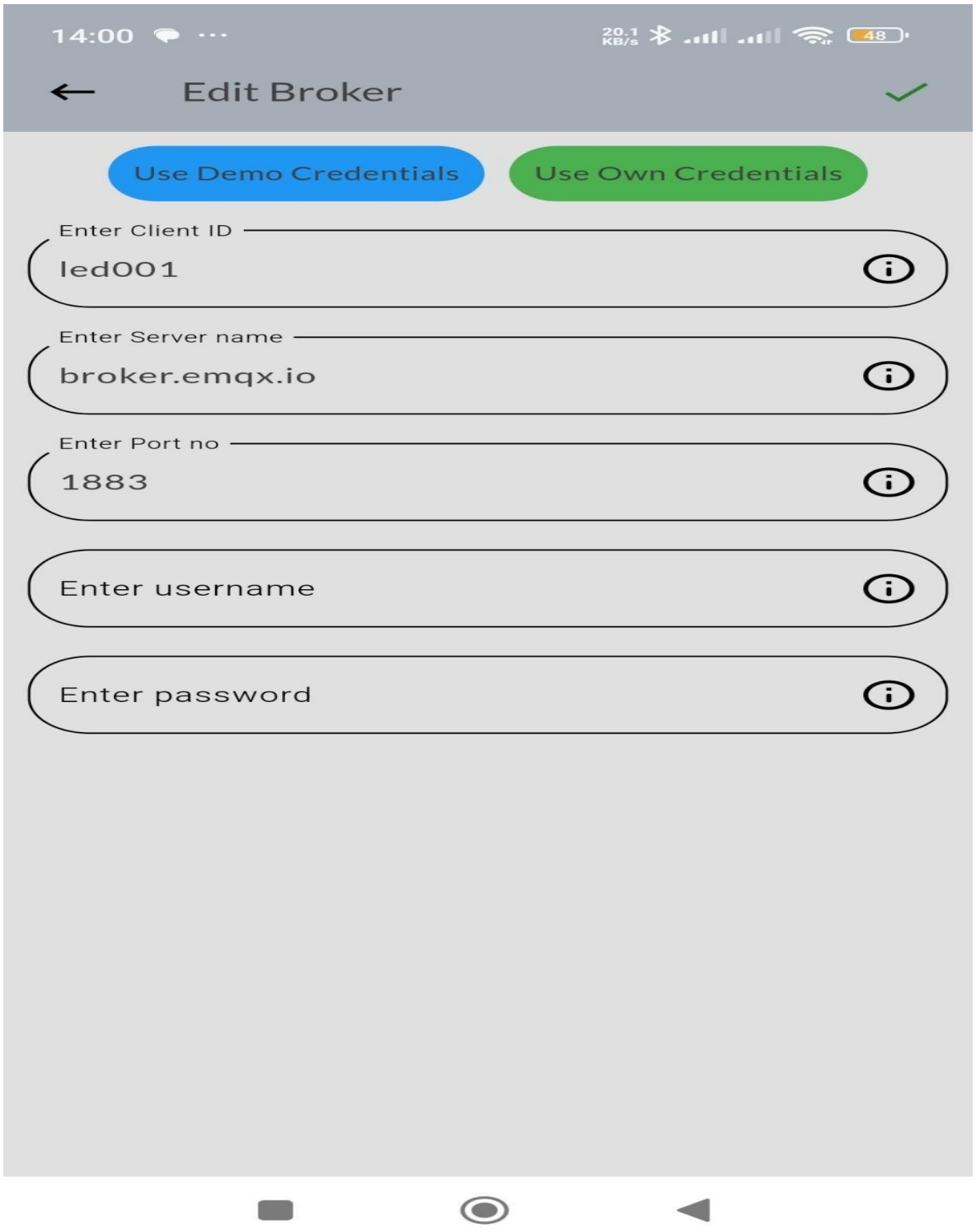
# Keep program running
client.loop_forever()
```

Instruction for MQTT Dashboard

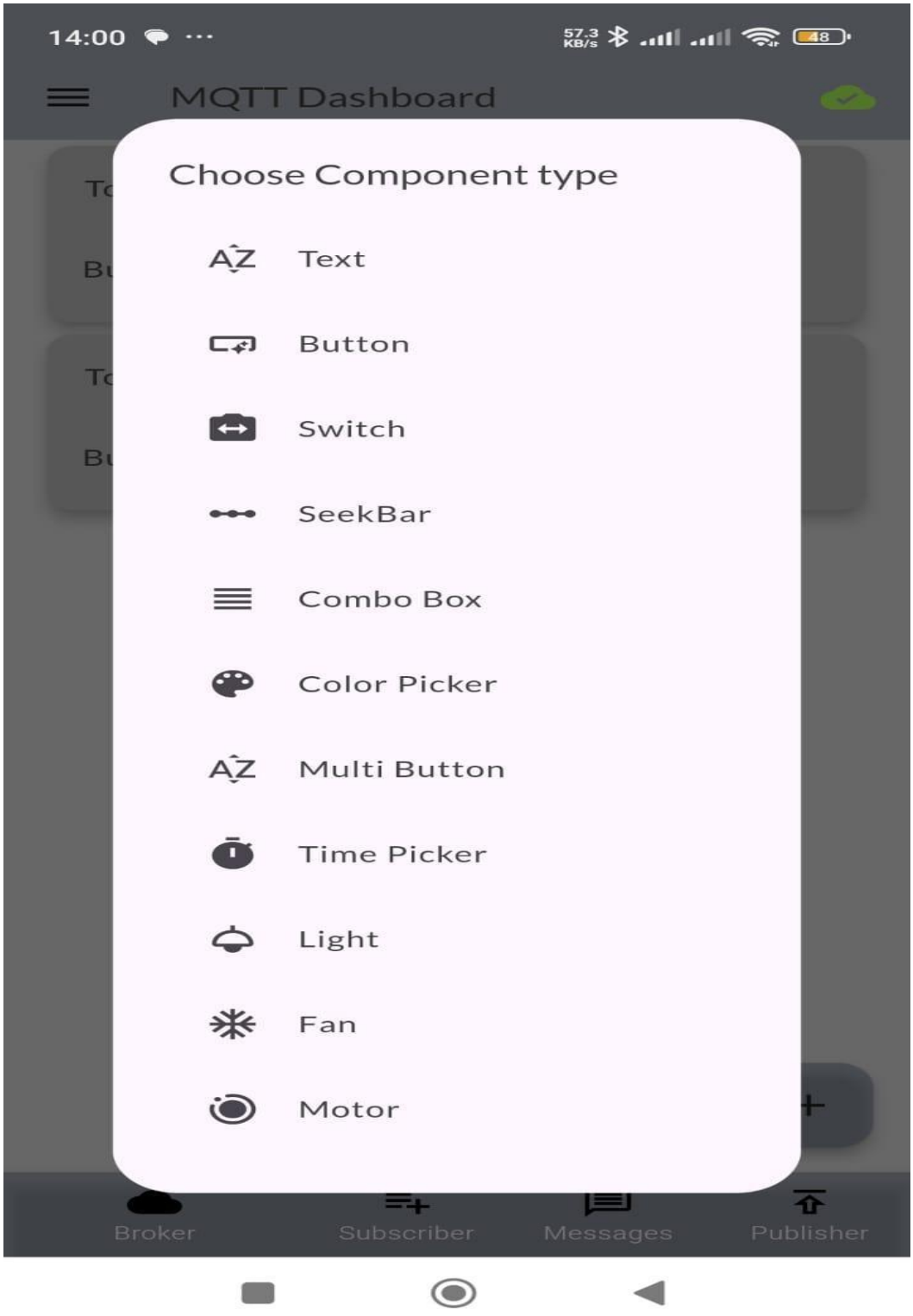
1. **Go to Play Store Download the application**

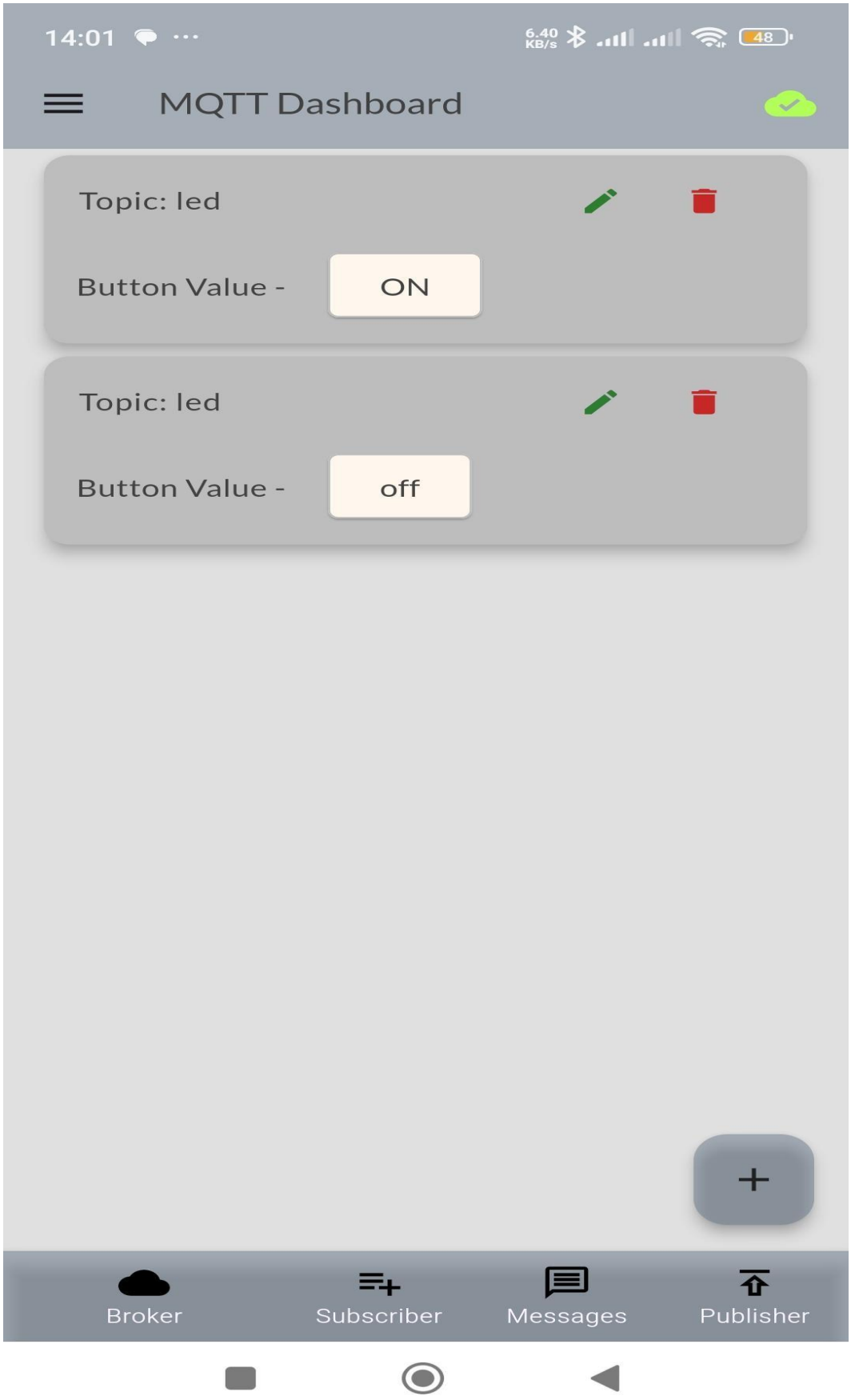


2. Add Your Broker Details



3. **Go to Publisher and go to the Button**




14:01
6.40 KB/s
48
MQTT Dashboard
Topic: led
Button Value -
ON
Topic: led
Button Value -
off
Broker
Subscriber
Messages
Publisher





To Run The Programme:-

1. python3 –m venv myenv
2. source myenv/bin/activate
3. pip install paho-mqtt
4. python3 test.py

